[1] Without loss of generality, assume f is bounded, namely, there exists constant $M>0$, such that $|f| \le M$. We know $\mu(\Omega)<\infty$, so we have.

$$\int_\Omega |f(x)| \mu(dx) \le \int_\Omega M \, \mu d(x) < \infty.$$

Thus f is μ- integrable.

[2] By Corollary 1.5.4, we know

$$\sum_{n=1}^{\infty} \int_\Omega |f_n| \, d\mu = \int_\Omega \sum_{n=1}^{\infty} |f_n| \, d\mu < \infty.$$

Hence $\int_\Omega \sum_{n=1}^{\infty} |f_n| \, d\mu < \infty$. So $\sum_{n=1}^{\infty} |f_n| < \infty$, a.e. by Thm 1.5.6

Since $\left|\sum_{k=1}^{n} f_k\right| \le \sum_{n=1}^{\infty} |f_n|$ and $\int_\Omega \sum_{k=1}^{\infty} |f_k| \, d\mu < \infty$,

By Theorem 1.5.8

$$\int_\Omega \sum_{k=1}^{\infty} f_k \, d\mu = \lim_{n\to\infty} \int_\Omega \sum_{k=1}^{n} f_k \, d\mu = \lim_{n\to\infty} \sum_{k=1}^{n} \int_\Omega f_k \, d\mu.$$

$$= \sum_{k=1}^{\infty} \int_\Omega f_k \, d\mu.$$

[3] By Theorem 1.5.7, we have.

$$\mu\left(\liminf_{n\to\infty} A_n\right) = \int_\Omega 1_{\left\{\liminf_{n\to\infty} A_n\right\}} \, d\mu \le \int_\Omega \liminf_{n\to\infty} 1_{A_n} \, d\mu \le \liminf_{n\to\infty} \int_\Omega 1_{A_n} \, d\mu.$$

$$= \liminf_{n\to\infty} \mu(A_n)$$

Note that $\Omega \setminus \left(\liminf_{n\to\infty} A_n\right) = \liminf_{n\to\infty} (\Omega \setminus A_n) = \liminf_{n\to\infty} A_n^c$

So $\mu(\Omega) - \mu\left(\limsup_{n\to\infty} A_n\right) = \mu\left(\liminf_{n\to\infty} A_n^c\right) \le \liminf_{n\to\infty} \mu(A_n^c)$

$$= \liminf_{n\to\infty} (\mu(\Omega) - \mu(A_n)) = \mu(\Omega) - \limsup_{n\to\infty} \mu(A_n)$$

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[4] Suppose that there exists some $A \in \mathcal{A}$ such that $\mu(A)>0$ but $\int_A f\,d\mu = 0$. Since $f>0$, a.e, we have $f \cdot 1_A \geq 0$, a.e.

Then $\int_A f\,d\mu = \int_\Omega f \cdot 1_A\,d\mu = 0$, which implies $f \cdot 1_A = 0$, a.e.

So $\mu(\{f>0\} \cap A) = 0$ and $\mu(\{f=0\} \cap A) = \mu(A) - \mu(\{f>0\} \cap A)$
$= \mu(A) > 0$,

which is a contradiction to $f>0$, a.e.

[5] It is sufficient to prove it when μ is finite. Let

$A_n = \{w : f(w) \geq g(w) + \frac{1}{n}\} \cap \{|g(w)| \leq n\}$. Then

$$\int_{A_n} g\,d\mu \geq \int_{A_n} f\,d\mu \geq \int_{A_n} g\,d\mu + \frac{1}{n}\mu(A_n)$$

But $\left|\int_{A_n} g\,d\mu\right| \leq \int_{A_n} |g|\,d\mu \leq n\mu(A_n) < \infty$, and thus $\mu(A_n) = 0$

Therefore $\mu(\bigcup_{n=1}^{\infty} A_n) = 0$; hence $\mu\{w : f(w) > g(w),\ g(w) \text{ finite}\} = 0$

taking $C_n := \{g = -\infty\} \cap \{f > -n\}$ we have $-\infty < C_n = \int_{C_n} g\,d\mu \geq \int_{C_n} f\,d\mu \geq -n\mu(C_n) > -\infty$, so $\mu(C_n) = 0$ and $\mu(\bigcup_n C_n) = 0$. Thus $g = f$ on $\{g = -\infty\}$.

Consequently, $f \leq g$, a.e. on $\{|g| < \infty\}$. Clearly $f \leq g$ on $\{g = +\infty\}$,

[6] Since $f \in L^1(\Omega)$, without loss of generality, assume that $f \geq 0$, we know $f < \infty$, a.e.

Define $A_n := \{w : \frac{1}{n} \leq f \leq n\}$, then we have $\lim_{n\to\infty} f \cdot 1_{A_n} = f \cdot 1_{\{f \neq 0\}}$ a.e.

By Theorem 1.5.8, we have

$$\lim_{n\to\infty} \int_\Omega f \cdot 1_{A_n}\,d\mu = \int_\Omega f \cdot 1_{\{f\neq 0\}}\,d\mu + \int_\Omega f \cdot 1_{\{f=0\}}\,d\mu$$
$$= \int_\Omega f\,d\mu.$$

Hence, for each $\varepsilon > 0$, there exists $N \in \mathbb{N}$, for $n \geq N$,

$$\left|\int_{A_n} f\,d\mu - \int_\Omega f\,d\mu\right| < \varepsilon.$$

On A_n, $\frac{1}{n} \leq f \leq n$, and $\frac{\mu(A_n)}{n} = \int_{A_n} \frac{1}{n}\,d\mu \leq \int_{A_n} f\,d\mu < \infty$

Therefore, $\mu(A_n) < \infty$.

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